

IV-to-Oral Antibiotic Switch (IVOS) in Orthopaedic Infections: Practice Patterns and Clinical Outcomes in a Malaysian District Hospital

Poster No: P04

NMRR ID: NMRR ID-25-01094-OWO

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KEY TAKE-HOME MESSAGE:

Early IVOS (24–96 hours) was commonly practiced (70.7%)
and significantly associated with **lower hospital readmission**
among orthopaedic patients.

1. INTRODUCTION

BACKGROUND

- IVOS is a core strategy in Antimicrobial Stewardship (AMS) to optimize therapy, reduce hospital length of stay (LOS), and minimize IV line complications.
- Guidelines recommend switching to oral antibiotics when patients are clinically stable (24–96 hours).
- Implementation is challenging in orthopaedic settings due to deep-seated infections and variable treatment durations.
- Limited Malaysian data link IVOS timing directly with post-discharge clinical outcomes like relapse and readmission.

OBJECTIVES

- Evaluate the prevalence and patterns of IVOS among orthopaedic patients.
- Determine the association between IVOS and clinical outcomes: treatment resolution, relapse, readmission and length of stay (LOS).
- Identify significant predictors of these clinical outcomes.

2. MATERIALS AND METHODS



Study Design

Retrospective cohort study



Setting

Ward 7
(Orthopaedic Ward),
Hospital Seri Manjung,
Perak



Study Population

Adult patients
(≥18 years) admitted
with orthopaedic
infections receiving
IV antibiotics
(Jun–Oct 2024)

n = 198



Data Sources

Bed Head Tickets
(BHT), Ward Census
and Pharmacy
Information System
(PhIS)

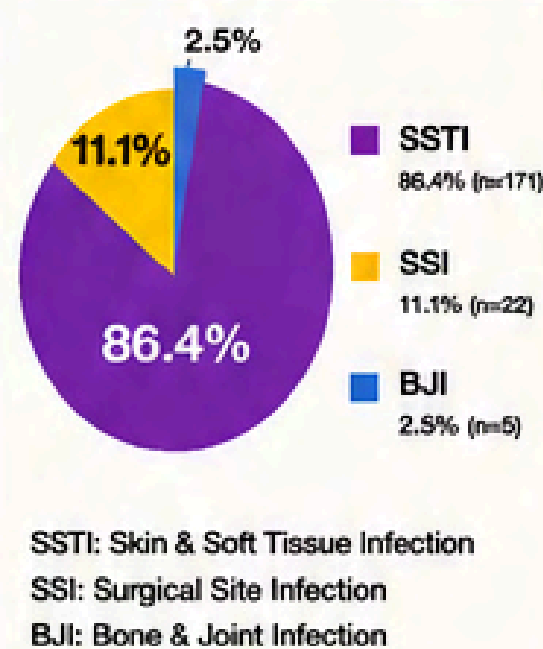


Data Analysis

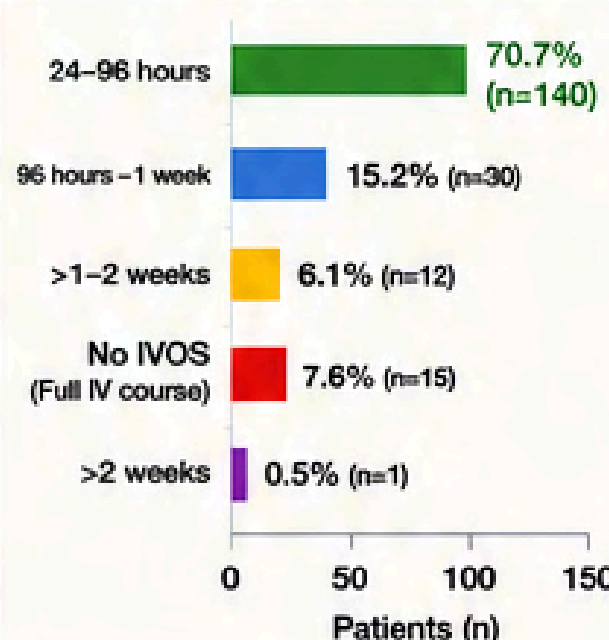
Analyzed using
SPSS v29.0.
Univariate followed by
multivariate logistic
and linear regression
to identify independent
predictors (p < 0.05).

3. RESULTS (n = 198)

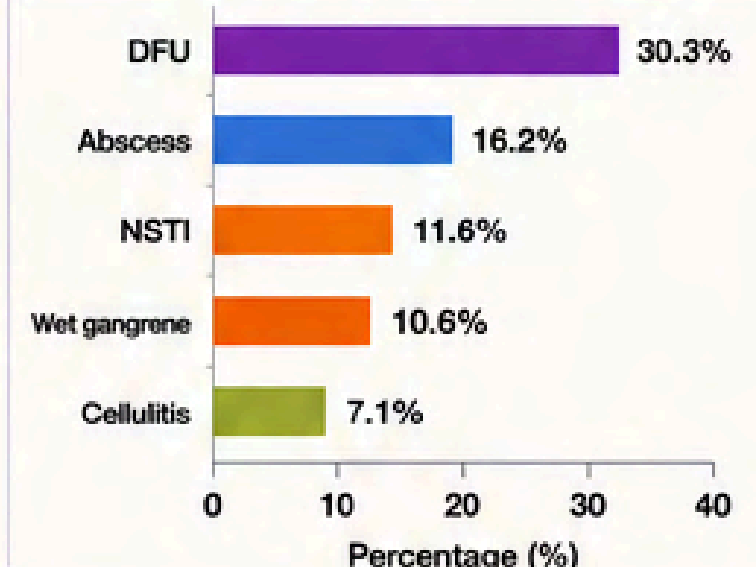
3.1 TYPES OF INFECTIONS



3.2 IVOS TIMING



3.3 TOP 5 DIAGNOSES



3.4 COMMON COMORBIDITIES

- Diabetes Mellitus
86.9% (n=172)
- Hypertension
70.2% (n=139)
- Cardiovascular Disease
22.2% (n=44)
- Renal Dysfunction
21.2% (n=42)

3.5 CLINICAL OUTCOMES

Treatment
Resolved



58.6%
(n=116)

Relapse
(Same Diagnosis)



29.3%
(n=58)

Reinfection
(Different Diagnosis)



14.6%
(n=29)

Readmission
(≤ 3 months)



52.2%
(n=103)



Median Length of Stay
3 days (IQR 2–6)

3.6. SIGNIFICANT PREDICTORS (Multivariate Analysis)

INCOMPLETE TREATMENT

- Diabetes Mellitus (Adj. OR = 6.76, p = 0.003)
- Surgical Site Infections vs SSSI (Adj. OR = 3.23, p = 0.020)

RELAPSE (Same Diagnosis)

- Cardiovascular Disease (↑ risk) (Adj. OR = 2.10, p = 0.090)
- Longer Oral Antibiotic Duration (↑ risk) (Adj. OR = 1.08/day, p = 0.047)

READMISSION (≤ 3 months)

- Hypertension (↑ risk) (Adj. OR = 4.30, p = 0.001)
- Early IVOS 24–96 hours (Protective Factor) (Adj. OR = 0.21, p = 0.014)

LONGER LENGTH OF STAY (Linear Regression)

- Renal Impairment (Adj. β = 0.77, p = 0.004)
- SSI/BJI vs SSSI (Adj. β = 0.37, p = 0.037)
- Longer IV Antibiotic Duration (Adj. β = 1.15, p < 0.001)

4. DISCUSSION / CONCLUSION

DISCUSSION & CONCLUSION

- Early IVOS within 24–96 hours is highly feasible in orthopaedic settings (70.7%) and significantly protects against hospital readmissions.
- Patient comorbidities (DM, HTN, CVD, Renal Impairment) are key determinants of clinical success and healthcare utilization.
- Prolonged oral antibiotic duration does not decrease relapse, highlighting that extended treatment reflects case complexity rather than improved efficacy.

RECOMMENDATIONS & IMPLICATIONS

- Institutionalize structured 48–96 hour IVOS screening checklists in daily orthopaedic ward rounds.
- Prioritize multidisciplinary comorbidity optimization (glycemic control, vascular care) alongside antibiotic management.
- Optimize IV duration to reduce unnecessary length of hospital stays.

5. ACKNOWLEDGEMENT



Special thanks to the Ministry of Health Malaysia (MOH), UiTM, University Malaya, and the staff of Hospital Seri Manjung & CRC Seri Manjung for supporting this research.

6. DECLARATION OF CONFLICT FOR ALL AUTHORS



All authors declare no conflicts of interest regarding this study.

7. REFERENCES

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